

Suvam Basak

Prime Minister’s Research Fellow at *Indian Institute of Technology Kanpur*

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Research Interests

Evaluate and simulate/emulate emerging networks and compute infrastructures in low Earth orbit (LEO) to understand design trade-offs and performance bottlenecks, and to explore possible design choices to address these limitations.

Education

Ph.D.	Indian Institute of Technology Kanpur (IIT-K) Department of Computer Science & Engineering Advisor: Dr. Amitangshu Pal & Dr. Debopam Bhattacharjee – CGPA: 8.75/10	Kanpur, Uttar Pradesh 2022 – Present
M.Tech.	University of Hyderabad (UoH) School of Computer and Information Sciences collaboration with IDRBT Thesis title: An Integrated and Standards-based Fog Computing Federation Framework Advisor: Prof. Satish Narayana Srirama – CGPA: 9.61/10	Hyderabad, Telangana 2020 – 2022
B.Tech.	West Bengal University of Technology (WBUT) Computer Science & Engineering, JIS College of Engineering CGPA: 8.6/10	Kolkata, West Bengal 2014 – 2018

Conference Publications

2025 USENIX ATC	LEOCraft: Towards Designing Performant LEO Networks – S. Basak, A. Pal, D. Bhattacharjee
2024 ACM IMC	CosmicDance: Measuring Low Earth Orbital Shifts due to Solar Radiations – S. Basak, A. Pal, D. Bhattacharjee
2022 IEEE Cloud	Fog Computing out of the Box with FogDEFT Framework: A Case Study – S. N. Srirama, S. Basak

Journal Publications

2023 ACM IJNM	Fog computing out of the box: Dynamic deployment of fog service containers with TOSCA – S. Basak, S. N. Srirama
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Workshop Publications

2025 SIGCOMM LEO-NET	An investigation of Starlink’s performance during the May ’24 solar superstorm – S. Basak, A. Pal, D. Bhattacharjee
2023 MobiCom LEO-NET	Exploring Low-Earth Orbit Network Design – S. Basak, A. Pal, D. Bhattacharjee

Talks

Jul 8, 2025 – Boston, USA	LEOCraft: Towards Designing Performant LEO Networks – USENIX ATC
Nov 4, 2024 – Madrid, Spain	CosmicDance: Measuring Low Earth Orbital Shifts due to Solar Radiation – ACM IMC

Teaching Experience

IIT Kanpur	Teaching Assistant, Department of CSE	
	- CS425 - Computer Networks	2025-26-II
	- ESC111/112 - Fundamentals of Computing - I/II	2025-26-I
	- ESC111/112 - Fundamentals of Computing - I/II	2024-25-II

- CS724 - Sensing, Communications and Networking for Smart Wireless Devices	2023-24-I
Teaching Assistant, eMasters in Financial Technology & Management	
- MBA977 - IT Platforms, Cloud Computing, and IoT	2023-24
Teaching Assistant, eMasters in Cyber Security	
- CS971 - Computer Networking II	2022-23
- CS963 - Computer Networking I	2022-23
Tutor, Department of CSE	
- ESC111/112 - Fundamentals of Computing - I/II	2022-23-II
- ESC111/112 - Fundamentals of Computing - I/II	2022-23-I
Tutorials and workshops	
- Advanced Python Programming	2024-26
- CSE-S208 - Data Structure Using Python	2023-24-II

UEM Kolkata

CSJM University (UIET)

Honors and Scholarships

Prime Minister's Research Fellowship (PMRF) – Cycle 11	2023 – 2026
Smart India Hackathon finalist (KL University, Guntur, Andhra Pradesh)	2017

Technical Skills

Programming:

Proficient in: C, Python — Familiar with: Flask, Verilog, Dart, Flutter, Solidity, JavaScript, Java

Tools & Softwares:

Gurobi Optimizer, Ansible, Docker, Docker Swarm, TOSCA, OpenStack, QEMU/KVM

Hardware:

Raspberry Pi, Arduino

Referees

IIT Kanpur

Dr. Amitangshu Pal, *Assistant Professor*

Department of Computer Science & Engineering

E-mail: amitangshu@cse.iitk.ac.in, **Web:** cse.iitk.ac.in/users/amitangshu

Microsoft Research

Dr. Debopam Bhattacharjee, *Senior Researcher*

E-mail: debopamb@microsoft.com, **Web:** bdebopam.github.io

University of Hyderabad

Prof. Satish Narayana Srirama, *Professor (Member IEEE)*

School of Computer and Information Sciences

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